

ex 1.13

$$Q_{in} = 35.4 \text{ nC}$$

$$\Phi_1 = 1250 \text{ N}\cdot\text{m}^2/\text{C}$$

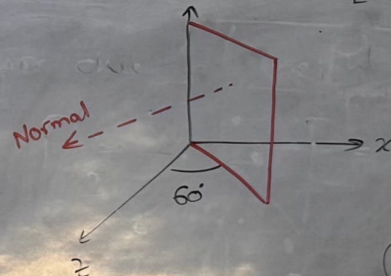
$$\Phi_E = \frac{Q_{in}}{\epsilon_0} = \frac{35.4 \times 10^{-9}}{8.85 \times 10^{-12}}$$

$$\Phi_E = 4000 \text{ N}\cdot\text{m}^2/\text{C} \text{ (total)}$$

$$\Phi_2 = \Phi_E - \Phi_1 = 4000 - 1250 \rightarrow \text{rounded}$$

$$\Phi_2 = 2750 \text{ (boxed)}$$

Q15



$$\vec{E} = -\hat{i} - 4\hat{j} + 2\hat{k}$$

Find the electric flux

$$\text{if } A = 2 \text{ m}^2$$

$$-2\cos 60^\circ \hat{i} = -\hat{i} \\ + 2\sin 60^\circ \hat{k} = \sqrt{3} \hat{k}$$

$$\vec{A} = -\hat{i} + \sqrt{3} \hat{k}$$

$$\begin{aligned} \Phi_E &= \vec{E} \cdot \vec{A} = (-\hat{i} - 4\hat{j} + 2\hat{k}) \cdot (-\hat{i} + \sqrt{3} \hat{k}) \\ &= 1 + 2\sqrt{3} = 4.46 \end{aligned}$$

Mass: Amount of matter in object.

Weight: Force of gravity on object

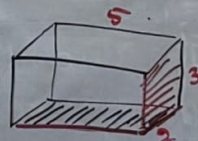
Volume: Amount of space of 3D object

Area: " " " " 2D "

Density: Mass per unit volume

$$1 \text{ g/cm}^3 = 1000 \text{ kg/m}^3$$

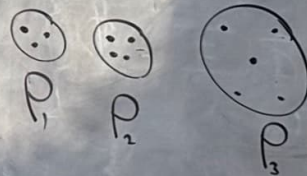
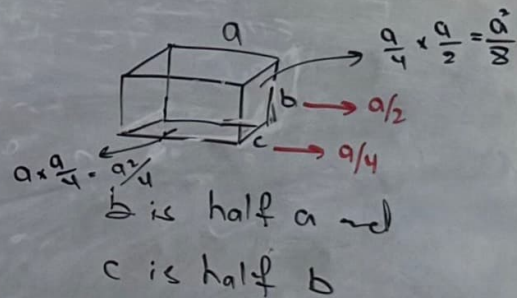
$$\frac{10^{-3} \text{ kg}}{10^{-6} \text{ m}^3}$$



$$V = 5 \times 2 \times 3 = 30$$

$$A = 5 \times 2 = 10$$

$$A = 2 \times 3 = 6$$



$$P_3 < P_1 < P_2 \text{ descending order}$$

$$P_2 > P_1 > P_3 \text{ Ascending order}$$

$$P_3 < P_2 > P_1$$

$$\rho = \frac{m}{V} \text{ kg/m}^3$$

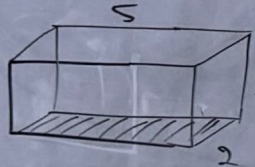
$$\rho_w = 1000 \text{ kg/m}^3$$

$$\rho_o = 900 \text{ kg/m}^3$$

① Pressure due to a force

$$P = \frac{F}{A} \quad \text{N/m}^2$$

$$P = 700 \text{ N/m}^2$$



$$V = 5 \times 2 \times 3 = 30$$

$$A = 5 \times 2 = 10$$

50 kg is placed on ground. Find the pressure on the ground

$$P = \frac{mg}{A} = \frac{50 \times 10}{10} = 50 \text{ N/m}^2$$

10^8 m^3

② Pressure due to a weight

$$P = \frac{W}{A} \quad \text{or} \quad P = \frac{m \times g}{A} \quad \text{or} \quad P = \frac{\rho V g}{A}$$

having density $\frac{1}{10} \text{ kg/m}^3$. Find the pressure

$$P = \frac{\rho V g}{A} = \frac{50 \times 30 \times 10}{10} = 1500 \text{ N/m}^2$$